

Report on Model 1 Performance – Logistic Regression

1. Overview

Logistic Regression was used as Model 1 to predict whether a customer would be classified as an existing or attrited customer in the Credit Card Customers (BankChurners) dataset.

The model used an 80/20 stratified train-test split with `random_state = 42`. Stratification was used to ensure that both the existing and attrite customer groups were represented in similar proportions in the training and testing data.

The model was configured as follows:

```
max_iter = 1000
```

```
class_weight = balanced
```

```
random_state = 42
```

The target variable was coded as:

Existing Customer = 0

Attrited Customer = 1

Categorical variables were converted using one-hot encoding, while the numerical variables were standardized before training.

2. Performance Results

The Logistic Regression model was evaluated using Accuracy, Precision, Recall, F1-score and ROC-AUC.

Metric	Result
Accuracy	0.8657
Precision	0.5538
Recall	0.8400
F1-score	0.6675
ROC-AUC	0.9421

The model achieved an accuracy of 86.57%, meaning that it correctly classified most of the customers in the test dataset.

The 84.00% recall is also important for this problem. It means that the model was able to identify 84% of the customers who were actually classified as attrited.

However, the 55.38% precision shows that not every customer predicted as attrited was actually attrited. In practice, this means that using the model to target customers for retention could result in some existing customers being incorrectly identified as at risk.

The F1-score of 0.6675 provides a balance between precision and recall, while the ROC-AUC of 0.9421 shows that the model was able to distinguish well between the two customer groups across different classification thresholds.

3. Statistical Evaluation Method

The model was tested on 20% of the dataset that was kept separate from the training data.

An 80/20 stratified split was used so that the imbalance between existing and attrited customers was maintained in both datasets.

The following measures were used to evaluate the model:

Accuracy: How many customers were classified correctly overall.

Precision: How often a customer predicted as attrited was actually attrited.

Recall: How many of the actual attrited customers the model managed to identify.

F1-score: A combined measure of precision and recall.

ROC-AUC: How well the model separates existing customers from attrited customers across different thresholds.

Using several measures rather than accuracy alone is important because attrited customers represent a smaller proportion of the dataset.

4. Why Logistic Regression Was Used

Logistic Regression was selected as the first model because it is relatively simple and easy to interpret.

This makes it useful as a starting point for the STADIOEquities problem. The purpose was not only to predict which customers may become dormant, but also to develop an approach where the relationship between customer characteristics and the prediction can be understood.

It also provides a useful baseline for comparing the performance of a more complex model such as XGBoost.

The results show that Logistic Regression was able to achieve 86.57% accuracy, 84.00% recall and a 0.9421 ROC-AUC on the public dataset. This indicates that even a relatively simple model can provide useful predictions of customer attrition.

At the same time, the precision of 55.38% shows that there is room for improvement, particularly in reducing incorrect attrition predictions.

5. Recommendation

Based on the results from the public dataset, Logistic Regression is useful as a baseline model, but the results should be considered alongside the performance of XGBoost.

XGBoost produced higher results across the five evaluation measures on the same test dataset. For this reason, it should also be tested when the project moves towards the STADIOEquities data.

However, the public dataset alone should not be used to make the final model decision. The models should first be tested on the client's actual data, and the business impact of false-positive and false-negative predictions should also be considered.

6. Discussion and Link to Part A

The results are broadly in line with the literature reviewed in Part A.

The studies reviewed showed that customer behaviour, transaction activity, inactivity, relationship characteristics and product usage can be useful when trying to predict customer churn or attrition.

A similar approach was followed in this project. The public dataset contains demographic, financial and behavioural information, and additional features were created from transaction activity and customer inactivity.

The literature also highlighted the importance of dealing with class imbalance. In this model, `class_weight = "balanced"` was used so that the minority attrited-customer class received greater consideration during training.

There is, however, an important difference between this public-dataset experiment and the actual STADIOEquities problem. The BankChurners dataset predicts customer attrition, while the STADIOEquities project is specifically concerned with customer dormancy.

These two concepts are related because both involve a reduction in customer engagement, but they are not exactly the same outcome. Therefore, the results from the BankChurners dataset should be treated as evidence that the modelling approach is worth testing, rather than as proof that it will produce the same results for STADIOEquities.

The next stage would therefore be to apply the same general approach to the client's data and determine which features and modelling techniques are most useful for predicting actual customer dormancy.

7. Related Code

The performance calculations are contained in:

`part_c_performance.ipynb`

The notebook includes:

Loading the BankChurners dataset.

Data preparation and feature engineering.

One-hot encoding.

Stratified train-test splitting.

Logistic Regression training.

Generating predictions.

Calculating the performance metrics.

Comparing the two models.

8. How to Use

Open `part_c_performance.ipynb` in Google Colab.

Upload BankChurners.csv and run the notebook from beginning to end.

The notebook will reproduce the Logistic Regression results reported in this document.

References

1. Peng, K., Peng, Y. and Li, W. (2023) 'Research on customer churn prediction and model interpretability analysis', PLOS ONE, 18(12), e0289724. Available at: <https://doi.org/10.1371/journal.pone.0289724> (Accessed: 28 September 2026).
2. Scikit-learn (2026) LogisticRegression. Available at: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html (Accessed: 28 September 2026).
3. XGBoost Developers (2026) *XGBoost Parameters*. Available at: <https://xgboost.readthedocs.io/en/stable/parameter.html> (Accessed: 28 September 2026).